

The Cold Component Keeps Its Collapse Mass: A Derived Dark-Sector Rule for a Λ -Tied MOND Law, Tested with Weak-Lensing Collapse Masses

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Abstract

The programme ties the MOND acceleration scale to the vacuum, $a_0 = \kappa c \sqrt{G \rho_\Lambda}$, with $\kappa = \frac{1}{2}$ fitted, not derived [Z26a]. Its best current version adds a cold fluid with the cosmic amount $\Omega_c h^2 = 0.12$. Inside bound systems the fluid supplies the law’s phantom density, and its cosmic share is written against the baryons a system holds *today*. That bookkeeping leaves baryon-depleted systems short. Twenty X-ray groups fall $1.4\times$ short at R_{500} and $1.9\times$ short inside R_{2500} , by an amount that tracks their lost baryons (Spearman $\rho = -0.96$). Seven X-ray-bright ellipticals fall short by a factor 1.9. *The rule*. The fluid is pressureless and conserved, so feedback cannot remove it: a bound system keeps the cold mass it collapsed with. The fluid first builds the law’s phantom, and any leftover stays as collapse debris with the collapse profile. No constant is added, but the collapse mass must be *measured*. Here it comes from weak lensing (Mandelbaum et al. 2016), which separates passive from star-forming central galaxies. *The tests*. With a colour-blind stellar-to-halo relation the rule breaks the most massive spirals (up to $+0.30$ dex in rotation speed). With the measured colour split:

- every star-forming spiral is left on the law;
- 16 passive early types with extended HI discs follow the law ($+0.026 \pm 0.085$ dex) and receive no debris;
- the X-ray ellipticals half-close ($+0.125$ dex, 1.0σ);
- the globular-cluster dispersions of 19 SLUGGS early types, which the law leaves $+0.080$ dex short (3.3σ), are fitted at $+0.007 \pm 0.017$ dex (0.4σ).

Added to the programme’s 48-gate harness, the rule moves SPARC by 0.0009 dex and the cold budget by at most 0.007 of Ω . One massive S0 in SPARC disagrees by $+0.14$ dex. Λ CDM on the same measured halos sits within about 2σ of the globular clusters, so these data do not favour the framework over Λ CDM.

1 The law and the dark sector it needs

The relation is $a_0 = \kappa c \sqrt{G \rho_\Lambda}$ with $\kappa = \frac{1}{2}$. Its form is forced and its coefficient is fitted [Z26a]; it predicts a_0 flat in redshift [Z26c]. Every number below is given on two footings: canonical, $a_0 \approx 9.36 \times 10^{-11} \text{ m s}^{-2}$; and alternative, $\approx 1.13 \times 10^{-10} \text{ m s}^{-2}$. Pairs are written canonical/alternative. The galaxy law is $g = \nu(g_N/a_0) g_N$ with the programme’s contract kernel ν_{mono} . It fits SPARC [LMS16] to 0.100 dex rms (lane CFG4).

The programme’s best candidate adds four ingredients (lanes CFG4, FG001):

- **T4:** a cold fluid with $\Omega_c h^2 = 0.12$, fitted as in Λ CDM, which behaves as CDM wherever the law is off;
- **T5:** the identity that inside a bound region the fluid *is* the law’s phantom, with dark mass $\max(M_{\text{ph}}, (\Omega_c/\Omega_b) M_b)$;
- **an edge:** the phantom ends at $x_e = 0.4$ of the turnaround radius;
- **hierarchical ownership:** accreted systems obey the isolated law of their infall baryons, and embedded ones are Newtonian.

With these it passes 42 of 48 harness gates (lane CFG19).

2 Where the as-written bookkeeping fails

T5 reproduces the X-COP clusters, whose baryon fraction is nearly cosmic: 0.946 of the hydrostatic mass (lane CFG4). It does not reproduce systems that have lost baryons:

- **Groups.** For twenty X-ray groups (Lovisari et al. 2015 [LRS15]; stars from the Kravtsov et al. 2018 relation [KVM18]) the hydrostatic mass exceeds B’s by $1.41/1.33$ at R_{500} . That is $1.8/1.5\sigma$ with a generous 20% hydrostatic allowance, and far more on the group-to-group error alone. Inside R_{2500} the excess is 1.88 (2.6σ). Across groups and clusters the shortfall tracks the baryon fraction, $\rho = -0.958$ ($p = 7 \times 10^{-18}$; lane CFG34).
- **X-ray ellipticals.** Seven X-ray-bright ellipticals (Humphrey et al. 2006 [H06]) need a factor $1.91/1.79$ more mass than the law gives ($1.7/1.6\sigma$ with the IMF floor), and the shortfall grows outward in all seven (lane CFG32).

3 The rule, derived

T4’s fluid is pressureless and its amount is conserved. Baryonic feedback is a pressure force on gas, so it moves gas and leaves the fluid. A bound system therefore keeps the cold mass it collapsed with, $M_c = (1 - f_b)M_{\text{coll}}$ with $f_b = \Omega_b/\Omega_m$. T5’s “cosmic share of today’s baryons” is only a floor.

T5’s identity says the fluid first *is* the phantom. Conservation says the leftover stays, and the law does not use it, so it keeps its collapse (CDM) profile. This is the *conservation form* (lane CFG35):

$$M_{\text{dark}}(< r) = M_{\text{ph}}(< r) + f_{\text{ex}}(1 - f_b) M_{\text{NFW}}(< r; M_{\text{coll}}), \quad f_{\text{ex}} = \max\left(0, 1 - \frac{M_{\text{ph,edge}}}{(1 - f_b)M_{\text{coll}}}\right). \quad (1)$$

Here M_{ph} is the law’s phantom from today’s baryons, and $M_{\text{ph,edge}}$ is its mass inside the edge. M_{NFW} has the Dutton & Macciò concentration [DM14]. For galaxies whose phantom already uses up their collapse mass, $f_{\text{ex}} = 0$ and the rule is the law.

No constant is added, but M_{coll} is history, not a prediction, so a non-circular test needs it measured. Before turnaround the fluid behaves as CDM, so the programme inherits Λ CDM’s collapse statistics. For single galaxies weak lensing measures the collapse mass directly.

4 Tests with measured collapse masses

Colour-blind masses break the spirals (lane CFG35). The Moster et al. relation [MNW13] puts every galaxy with $\log M_* \approx 11.4$ in a $\sim 10^{14} M_\odot$ collapse. The rule then closes the X-ray ellipticals ($-0.013/-0.024$ dex). It also raises the outer rotation speed of the most massive SPARC spirals by up to $+0.30$ dex (UGC 2885), and their rotation curves follow the law. The rule cannot take a colour-blind relation.

Measured colour-split masses (lane CFG36). Mandelbaum et al. 2016 [M16] measured the mean halo mass of central galaxies by weak lensing, separately for passive and star-forming systems. Their Table 3 is transcribed with provenance in the repository. At fixed stellar mass, passive centrals sit in halos $3-7\times$ heavier. With the red relation for early types and the blue one for late types:

- every late-type SPARC spiral in the table’s range has $f_{\text{ex}} = 0$;
- the X-ray ellipticals half-close ($+0.125/+0.122$ dex, 1.0σ);
- the one massive S0 in SPARC, UGC 2487, would rotate $+0.14$ dex faster than observed.

Passive disks (lane CFG37). Sixteen ATLAS^{3D} early types with HI circular velocities at 8–28 kpc (den Heijer et al. 2015 [dH15]) follow the law at $+0.026 \pm 0.085$ dex (0.3σ). With $\log M_* = 10.3-10.9$ their collapse masses are used up by the phantom ($f_{\text{ex}} = 0$ in all sixteen), so the rule predicts the same thing and passes too.

Massive early types (lane CFG38). The SLUGGS survey measured globular-cluster dispersion profiles of 19 early types out to 5–15 effective radii (Forbes et al. 2017 [F17]). The law leaves them $+0.080 \pm 0.024$ dex short (3.3σ), worst in the most massive (M87 $+0.28$). With the red collapse masses the rule adds debris to the ten galaxies above $\log M_* \approx 11.2$ and fits the sample at $+0.007 \pm 0.017$ dex (0.4σ). M87 moves to $+0.08$, and the rise of the offset with mass disappears.

sample	the law	the rule (measured masses)	$f_{\text{ex}} > 0$
late-type SPARC spirals (CFG36)	fits	identical	none
16 passive HI discs (CFG37)	$+0.026$ (0.3σ)	identical	none
19 SLUGGS early types (CFG38)	$+0.080$ (3.3σ)	$+0.007$ (0.4σ)	10 of 19
7 X-ray ellipticals (CFG36)	$+0.28$	$+0.125$ (1.0σ)	6 of 7
UGC 2487 (S0, CFG36)	fits	$+0.14$ dex too fast	yes

The harness (lane CFG39). The rule changes only systems with $f_{\text{ex}} > 0$:

- **SPARC:** the rms moves from 0.1003 to 0.1012 dex; only UGC 2487 is touched.
- **Cold budget:** it gains at most 0.007 of Ω and stays within Ω_c .
- **KiDS:** the isolated lenses [Bro21] carry no debris.
- **X-COP and the Bullet:** they cannot be re-scored without circularity. The rule can only raise their dark mass.

The score therefore stays at 42/48, inferred rather than established: the lane’s declared test failed on an artefact of its own threshold scan, which is recorded.

5 What is not claimed

- **The data do not favour the framework over Λ CDM.** Λ CDM on the same measured halos (stars plus the full halo, no phantom) sits at -0.037 ± 0.017 dex on the globular clusters (2.1σ ; a post-hoc comparison). It has its own modelling freedom: contraction, anisotropy, the IMF.
- **The rule is not a prediction of individual collapse masses.** It inherits the stellar-to-halo relation’s systematics: stellar-mass conventions, and centrals versus satellites (M87 is a cluster central; NGC 4374, 4365 and 4649 are Virgo members).
- **Groups and clusters are not scored.** Every collapse-mass estimate available here comes from their own mass.
- **One galaxy disagrees.** UGC 2487 is one massive S0 with a measured rotation curve.

6 Tests that decide it

The rule makes a sharp prediction where it differs from the law. **Massive passive disks** ($\log M_* > 11.2$) **with extended HI or tracer kinematics must rotate faster than the law at large radii, by the amount their colour’s collapse mass implies. Star-forming disks of the same stellar mass must not.** That sample is small but observable. The rule also inherits the programme’s standing tests: Gaia DR4 wide binaries (2 December 2026) and a_0 at $z \approx 2.5$.

7 Reproducibility

Every lane is a committed script in `campaign_fresh_gravity/` (CFG32, CFG34–CFG39). Each run has declared hypotheses, controls that reproduce committed numbers, and a mutation control that must fail. Changes made after a mutation run and before the main run are disclosed in each script. The new data (Mandelbaum et al. Table 3; den Heijer et al. Table 1) are in `real_research/data/` with provenance and hashes. `PAPER36_audit.py` re-reads every number quoted here from the committed outputs.

References

- [Bro21] Brouwer, M. M. et al. 2021, A&A 650, A113 (KiDS-1000 lensing RAR).
- [dH15] den Heijer, M. et al. 2015, A&A 581, A98.
- [DM14] Dutton, A. A. & Macciò, A. V. 2014, MNRAS 441, 3359.
- [F17] Forbes, D. A. et al. 2017, AJ 153, 114 (SLUGGS).
- [H06] Humphrey, P. J. et al. 2006, ApJ 646, 899.
- [KVM18] Kravtsov, A. V., Vikhlinin, A. A. & Meshcheryakov, A. V. 2018, Astronomy Letters 44, 8.
- [LMS16] Lelli, F., McGaugh, S. S. & Schombert, J. M. 2016, AJ 152, 157 (SPARC).
- [LRS15] Lovisari, L., Reiprich, T. H. & Schellenberger, G. 2015, A&A 573, A118.
- [M16] Mandelbaum, R., Wang, W., Zu, Y., White, S., Henriques, B. & More, S. 2016, MNRAS 457, 3200.
- [MNW13] Moster, B. P., Naab, T. & White, S. D. M. 2013, MNRAS 428, 3121.
- [Z26a] Zimmerman, C. P. 2026, The coefficient of the a_0 – Λ relation: a zero-mode theorem for local MOND actions, DOI 10.5281/zenodo.22559892.
- [Z26c] Zimmerman, C. P. 2026, Does the MOND acceleration scale evolve? A pre-registered decisive measurement (PAPER7 v3), DOI 10.5281/zenodo.22833314.